

R E V I E W

Role of the leukocyte-rich PRP components in tissue repair and regeneration

SHEILA VERONESE¹, ANDREA SBARBATI¹

¹*Department of Neuroscience, Biomedicine, and Movement Sciences, University of Verona, Verona, Italy*

ABSTRACT

Background: Platelet-rich plasma (PRP) is an autologous plasma derivative, widely used in regenerative medicine, and recently applied to aesthetic medicine. In the literature, many types of PRP are described, and the presence and amount of leukocytes are the subject of discussion, particularly for the so-called leukocyte-rich PRP (LR-PRP).

Aim: This narrative review aims to analyse the different compositions of LR-PRP to highlight how their characterization can affect the results of product application.

Methods: A literature search was conducted in the PubMed database for relevant studies published up to March 2026. Only articles that debated biological aspects were considered.

Results: The research yielded 105 articles, and cross-references, similar articles, and citing articles were analysed.

Conclusions: The review clarified the roles of the different components of the LR-PRP. A high concentration of platelets and mononuclear cells is desirable for tissue repair and regeneration. However, a possible healing role of granulocytes has emerged, while the presence of damaged erythrocytes must not be underestimated.

Key words: LR-PRP, Platelet Concentration, Leukocytes, Mononuclear Cells, Neutrophils, Erythrocytes



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Correspondence: Sheila Veronese / Department of Neuroscience, Biomedicine, and Movement Sciences, University of Verona, P.le L.A.Scuro 10, 37134 Verona, Italy / E-mail: sheila.veronese@univr.it

Introduction

The term “platelet-rich plasma” (PRP) was first used by Kingsley in 1954¹. In his work, Kingsley identified an antagonist of Factor VI in PRP for the regulation of the coagulation speed. Consequently, PRP was initially utilized as a transfusion product for the treatment of thrombocytopenia^{2,3}. Subsequently, in the 1970s and 1980s, its use expanded because, thanks to the growth factors it contains, it promotes tissue repair. First, it was applied in the oral and maxillofacial surgery⁴⁻⁶. Then, its use was extended to cardiovascular surgeries to facilitate postoperative healing, protect against infection, and promote angiogenesis and arteriogenesis in the infarcts⁷⁻¹⁰. Finally, it spread into regenerative medicine, starting in sports medicine and orthopaedics for tendon, ligament, and cartilage repair¹¹⁻¹³. Nowadays, PRP is finding many uses in aesthetic medicine, as it is applied in facial rejuvenation, acne scar healing, hair restoration, and other aesthetic applications, alone or in combination with other treatments¹⁴⁻¹⁶.

Its multiple uses are related to its composition. In fact, PRP is an autologous plasma derivative that contains a platelet concentration above baseline. However, the simple term “PRP” is insufficient, as there are different methodologies for obtaining it from blood, and the final product can vary widely^{17,18}. Moreover, in the literature, the number of terms used is very large¹⁸⁻²¹. If the PRP contains only platelets, it is called “pure PRP,” often abbreviated to PPP. If the preparation also contains leukocytes, it is called leukocyte-rich PRP (LR-PRP or L-PRP) or leukocyte-poor PRP (LP-PRP, P-PRP, or PRP), depending on the amount of leukocytes present.

The main difference among the three preparations seems to be that PPP and LP-PRP present chemotactic and proliferative properties. So, they promote matrix anabolism, whereas LR-PRP has proinflammatory characteristics that are undesirable²². In reality, the literature is full of studies comparing the different preparations. PPP seems to be the preferable solution for bone repair²³, in knee osteoarthritis²⁴, in tendon repair^{25,26}, and in joint arthritis²⁷. Other authors have underlined the superiority of LP-PRP with respect to LR-PRP in: knee

osteoarthritis²⁸, intervertebral disc degeneration²⁹, osteochondral lesions of the talus³⁰, intra-articular injury³¹, and tendinopathy³². Of particular interest is the finding from Chen et al.³³, who reported superior functional recovery with LP-PRP compared with LR-PRP in osteoarthritis, but similar pain relief with the two products. However, this last result is not supported by Li et al.³⁴, who found a lower temporary pain rate in LP-PRP than in LR-PRP. From these studies, it seems clear that leukocytes are not beneficial in the healing phases of the various pathologies analyzed. Nonetheless, other studies have found no differences in results when applying the two solutions³⁵⁻³⁹, and others have underlined that LR-PRP appears to offer major benefits⁴⁰⁻⁴⁶.

Indeed, Cavallo et al.³⁹ in 2014 had already noted that different PRPs may be necessary for different problems. This concept has recently been taken up by Yu et al.⁴⁷, who recommended LR-PRP in the early stage of osteoarthritis and LP-PRP in advanced osteoarthritis, and by Peng et al.⁴⁸, who suggested using LR-PRP in the early stage of healing after rotator cuff injury and LP-PRP in the later stages. Cavallo et al.³⁹ summarized these considerations, stating that at different stages of a disease, it is necessary to select the appropriate PRP formulation.

This study stems from this last statement, and its aim is to analyse the different compositions of LR-PRP, in particular to highlight how the characterization of the leukocytes and the other components present in the preparation can affect the results of the product application. This could explain the contrasting results reported in the literature.

Materials and methods

This study is a narrative review aimed at clarifying the roles of the different components of the LR-PRP in the healing process. A literature search was undertaken in March 2026 in PubMed (United States National Library of Medicine, Bethesda, MD) of articles related to LR-PRP. Given the purpose of the study, it was not considered necessary to utilize multiple databases for research.

Search terms

The terms used for the search were: “PRP”, “LR-PRP”, “L-PRP”, “leukocytes”, “platelets”, “white blood cells”, “monocytes”, “lymphocytes”, “mononuclear cells”, “neutrophils”, “eosinophils”, “basophils”, “granulocytes”, “erythrocytes”, and “red blood cells”. The terms were searched alone and in combinations. The only ineligibility criterion was the non-English language of the studies. The eligibility criteria were: studies with full-text availability. Cross-references, similar articles, and citing articles were subsequently analyzed. Clinical studies on the application of PRP that did not address biological considerations were excluded.

Results

Search results

Pubmed research with the single term LR-PRP identified 108 articles. Two articles were removed according to the ineligibility criterion. Another article did not meet the eligibility criteria and was excluded from the analysis. For all the 105 remaining articles, cross-references, similar articles, and citing articles were analyzed.

Discussion

First result: the absence of a standard

The first debatable result concerns PRP in general. A product with a high platelet concentration is preferred, as it is considered more effective from a therapeutic standpoint; however, there is no standard value defining the minimum platelet concentration required for the preparation to be considered “therapeutic”. The baseline range is 150,000–450,000 platelets/ μL = $150\text{--}450 \times 10^9$ platelets/L⁴⁹.

Mazzucco et al.⁵⁰ reported that a PRP with a platelet count greater than $200 \times 10^3/\mu\text{L}$ ($200 \times 10^9/\text{L}$) was therapeutic, but this value may fall within the normal range. Giusti et al.⁵¹ found PRP containing 500×10^3 platelets/ μL ($500 \times 10^9/\text{L}$) to be more therapeutic, which, in many subjects, is just above

baseline levels. Marx⁵² (2001), Bansal et al.⁵³, and Weibrich et al.⁵⁴ preferred a value around $1,000 \times 10^9$ platelets/L, while Graziani et al.⁵⁵ preferred $1,500 \times 10^9$ platelets/L.

What emerged from many studies is that too-low concentrations may lead to suboptimal effects, while too-high concentrations may result in paradoxical inhibitory effects^{51,54–56}. Thus, an ideal processing should yield a PRP containing $1,000\text{--}1,200 \times 10^9$ platelets/L²¹.

LR-PRP: Which characteristics?

LR-PRP is obtained using the method generally referred to as the “Buffy coat” method¹⁸, and various commercial kits or extraction procedures are available. Different studies in the literature reported varied results regarding preparations, both in terms of platelet concentrations and leukocyte concentrations, as well as the presence of red blood cells⁵⁷.

Published data on devices already on the market often show conflicting results. In the review by Oudelaar et al.⁵⁷, for instance, the GPS III (Zimmer Biomet), SmartPrep (Harvest Technologies), and Magellan (Arteriocyte) systems were indicated as the best devices for LR-PRP extraction. The GPS III and SmartPrep systems reported a platelet count of $1,000 - 1,500 \times 10^3/\mu\text{L}$ ($1,000 - 1,500 \times 10^9/\text{L}$). The same review, however, reported two studies on the Magellan system that yielded very different final platelet concentrations, each lower than those obtained with the GPS III and SmartPrep systems. It should be noted, however, that in a 2011 article by Castillo et al.⁵⁸, a lower platelet count was reported for the GPS III device, at $566.2 \pm 292.6 \times 10^9$ platelets/L. This demonstrates considerable variability in published data, attributable to many factors, primarily the centrifugation systems used¹⁸.

It is clear that the type of biological response achievable in clinical practice depends on the leukocytes present. Many studies reported only leukocyte counts. However, if only the leukocyte count were considered, taking into account, for instance, the already mentioned GPS III system device, different data would still be obtained across studies. For example, Oudelaar et al.⁵⁷ reported a leukocyte count of $20\text{--}30 \times 10^9/\text{L}$, while Castillo et al.⁵⁸ reported a value of $34.4 \pm 13.6 \times 10^9$ leukocytes/L.

As already stated, the inclusion of leukocytes in PRP has been and remains a source of debate. Indeed, their presence can provide benefits or cause adverse events^{57,59}. On the one hand, Dohan et al.¹⁷ and Moojen et al.⁶⁰ highlighted the role of leukocytes in tissue remodeling, and in increasing the antibacterial properties of the preparation and immunological resistance in treated tissues, and Davis et al.⁶¹, Denapoli et al.⁶², Kobayashi et al.⁶³, and Zimmermann et al.⁶⁴ highlighted the correlation between the number of leukocytes and the number of growth factors present. On the other hand, other authors have highlighted negative catabolic and inflammatory effects on tissues, linked to the release of proinflammatory cytokines by leukocytes^{22,24,26,31,39,65–68}. In detail, among the various pro-inflammatory factors that are upregulated, interleukin-1 β (IL-1 β) and interleukin-8 (IL-8) were most frequently reported. Interestingly, Mariani et al.⁶⁹ found differences in the *in vivo* behavior of the obtained PRP compared with its *in vitro* behavior. They reported that the pro-inflammatory upregulation, observed *in vitro*, was not confirmed *in vivo*.

The different data reported for the individual devices and the different kinds of response that a LR-PRP can draw attention to the quality of the product obtained. Authors such as Sundman et al.⁷⁰, Mautner et al.²¹, and Dohan et al.¹⁷ highlighted how the quality assessment of LR-PRP depends on the type of leukocytes present.

Leukocytes characterization

Leukocytes derive from lymphoid and myeloid cells, the two main cell lineages derived from hematopoietic stem cells in the bone marrow. Lymphoid cells give rise to lymphocytes, while myeloid cells give rise to other types of leukocytes, namely monocytes, neutrophils, basophils, and eosinophils⁴⁶.

LR-PRP seems characterized by a reversal of the normal leukocyte count, with a reversal of the percentages of the two most representative leukocytes in the blood (neutrophils and lymphocytes). Neutrophils tend to go from 60% to 30%, while lymphocytes go from 30% to 55%^{65,71–75}. In general, LR-PRP presents a high number of monocytes and lymphocytes (mononuclear cells), which appears to be optimal for

promoting tissue regeneration and modulating the inflammatory response, and a low number of neutrophils, eosinophils, and basophils (granulocytes).

Concerning monocytes, once injected into the tissue, they transform into macrophages⁷⁶, which phenotypically express in M1 or M2 type. M1 acts as a pro-inflammatory cell type, while M2 acts as an anti-inflammatory cell type. Macrophages can polarize into the two subtypes depending on the microenvironment⁷⁷. The M1 cells intervene in the first phase of healing, triggering a pathogen-killing mechanism^{78,79}, and initiating the formation of new blood vessels (angiogenesis) through the production of vascular endothelial growth factor (VEGF) and fibroblast growth factor (FGF)⁸⁰. In the second phase of healing, M1 cells switch to M2 cells^{81,82}. The latter have a paramount role in tissue repair^{83,84}, as they produce extracellular matrix (ECM), collagen, growth factors, and stimulate angiogenesis and fibroplasia⁸⁵. These functions permit them to drive wound closure and tissue regeneration^{81,86}, and resolve inflammation through the production of interleukin-10 (IL-10)^{79,81}.

Lymphocytes comprise B and T cells, helper cells, and natural killer cells⁷⁵. They contribute to the immunological and regulatory components of the preparation and the control of the local immune response, ensuring that the healing process does not become chronic or excessively inflammatory⁸⁷. Moreover, they drive monocyte-to-macrophage differentiation^{88,89} and strengthen macrophage polarization⁹⁰. This means that, together with monocytes, lymphocytes create a microenvironment favorable to the survival of local stem cells and the repair of damaged tissues^{71–73,80}.

In LR-PRP, the reduction of granulocytes (neutrophils, eosinophils, and basophils), and particularly of neutrophils, which are pro-inflammatory cells, appears to be beneficial. A high eosinophil count increases the risk of local allergic reactions or nonspecific inflammation^{71,91}. However, it must be noted that their role in modulating T-helper cell differentiation and, thus, in tissue regeneration has already been proven⁹². The number of basophils generally present in LR-PRP is not counted because it is so low that it does not significantly affect the preparation's effect on tissues⁷². However, as for eosinophils, their role is questionable because they can be both

pro-inflammatory⁹³ and capable of releasing platelet-activating factor (PAF), which, in turn, further enhances platelet activation and growth factor release^{71,94–96}.

Neutrophils deserve an in-depth analysis. They are well known as pro-inflammatory mediators and as promoters of catabolism. In fact, they release inflammatory cytokines and matrix metalloproteinases⁹⁷. They are considered responsible for the acute inflammatory response observed, for instance, in tendons 5 days after injection⁶⁷. Their massive presence in a PRP can lead to collagen degradation and fibrosis. Zhou and Wang reported a high collagen type III to type I ratio in PRP rich in neutrophils, which led to tendon fibrosis and damage⁹⁸. Nevertheless, their role has been re-evaluated, as they seem crucial to the healing process^{99–101}. Moreover, neutrophils convert arachidonic acid to leukotriene A4, one of the main actors in the inflammation cascade¹⁰², which is then converted to lipoxin A4 through interaction with activated platelets¹⁰³. Lipoxin A4 is a potent anti-inflammatory protein that is able to inhibit the proinflammatory action of neutrophils^{80,103,104}. Finally, Marwick et al. proved that neutrophils induce macrophage anti-inflammatory reprogramming, favouring the healing phase and regenerative process¹⁰⁵. These considerations must be applied to the LR-PRP solution. It seems advisable to prefer preparations with a low neutrophil count to reduce or control, particularly to reduce the acute initial inflammatory response. In addition, knowing the number of neutrophils can help predict and manage the type of inflammatory response, and this number can serve as a parameter for evaluating the LR-PRP obtained. It must be specified that no such characterization appears to exist in the literature for commercial devices. Some studies characterized the LR-PRP obtained, but the reported neutrophil concentrations were extremely variable. For instance, Ziegler et al.⁴⁰ reported 9.1×10^9 neutrophils/L, Yaradilmis et al.⁴⁶ reported $23.133 \pm 7.3 \times 10^9$ neutrophils/L, while Kobayashi et al.⁶³ reported $5.4 \pm 3.0 \times 10^9$ neutrophils/L. Therefore, comparing devices solely on the basis of neutrophil counts appears impossible or, even, inadequate.

How to evaluate an LR-PRP: possible solution

It seems clear that the evaluation of the quality of a LR-PRP must depend on all its components. The reversal in neutrophil and lymphocyte percentages,

and, in general, the leukocyte count, could be attributed to the processing systems' ability to perform selective filtration. This filtration appears to be related to the specific gravity of individual cells, which is equal to:

- Platelets: 1.040 – 1.060 g/ml;
- Monocytes: 1.058 – 1.068 g/ml;
- Lymphocytes: 1.065 – 1.075 g/ml;
- Neutrophils, Eosinophils, and Basophils: 1.080 – 1.095 g/ml;
- Erythrocytes: 1.090 – 1.100 g/ml.

In this context, filtration capacity could be a parameter to evaluate the efficiency of different PRP treatment systems: the greater the separation, the better the system. Good processing should be able to retain the majority of platelets and mononuclear cells. By limiting the amount of buffy coat collected, it should be possible to regulate the number of granulocytes and erythrocytes present in the preparation, and consequently modulate the inflammatory response.

Erythrocytes: not to be underestimated

The presence of red blood cells in the preparation should not be underestimated. According to Tajima et al.¹⁰⁶ (2017), haemoglobin exacerbates synovitis, a crucial factor in the development of osteoarthritis. During PRP extraction, the erythrocyte membrane can be damaged, leading to the release of plasma-free haemoglobin (PFH), hemein, and iron¹⁰⁷, which are cytotoxic. These products can cause oxidative stress, nitric oxide loss, inflammation activation, and immunosuppression. This means that the final effect can be tissue injury rather than tissue repair or regeneration⁹¹. Therefore, a PRP with a low red blood cell concentration is desirable⁷².

Conclusions

Considering what has been expressed in the literature, it is clear that PRP preparations must be evaluated as a whole, taking into account all their components to understand how the different quantities of these may influence tissue response. Moreover, this

review highlighted a possible system for evaluating the numerous devices available on the market for PRP preparation, namely their ability to perform selective filtration.

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